



SoK: Blockchain Consensus in the Quantum Age

Aleck Nash

aleck.nash@my.utsa.edu

University of Texas at San Antonio

Henry Chimal-Dzul

henry.chimal-dzul@utsa.edu

University of Texas at San Antonio

Christian Eduardo Terron Garcia

ChristianEduardo.TerronGarcia@my.utsa.edu

University of Texas at San Antonio

Kim-Kwang Raymond Choo

raymond.choo@fulbrightmail.org

University of Texas at San Antonio

Abstract

Consensus protocols are an important building block in blockchain and blockchain-based systems. The recent focus in developing practical quantum computers reinforces the importance of designing quantum-resistant cryptographic protocols, and in the context of this paper quantum-resistant consensus protocols. In this paper, we systematically review the extant literature on quantum-resistant consensus protocols published between 2019 and 2024. As part of the review, we identify a number of limitations and challenges and discuss potential future research directions.

CCS Concepts

• **Security and privacy** → **Pseudonymity, anonymity and untraceability.**

Keywords

Consensus Protocol, Post-Quantum Cryptography, Lattice-based Cryptography, Multivariate Cryptography

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1 Introduction

Consensus protocols play a key role in blockchain applications in achieving agreement between network nodes to ensure the accuracy and consistency of distributed information. Table 1 describes existing popular consensus protocols, whose security is generally based on computationally hard problems. Among these protocols, Proof of Work (PoW) is widely used and relies on competition among miners, who attempt to solve a computational puzzle by finding a hash that satisfies a network-defined difficulty threshold.

Advances in quantum computing as theoretically demonstrated by Shor [32] and Grover [21], however, pose a significant risk to such classical consensus protocols. Grover's algorithm, for example, provides a quadratic speed-up for unstructured search problems, allowing quantum computers to find hash puzzle solutions faster

Table 1: Popular Consensus Protocols in Blockchain

Protocol	Description
Proof of Work (PoW)	Participants compete by solving computationally intensive cryptographic puzzles; first to solve validates the next block.
Proof of Stake (PoS)	Validators are selected proportionally based on the cryptocurrency amount they stake as collateral; higher stakes increase validation chances.
Secret Leader Election	Protocol selects randomly and confidentially a block proposer, protecting the proposer from targeted attacks until the block is published.
Proof of Luck	Consensus achieved through secure hardware-generated randomness (trusted execution environments), randomly selecting block proposers.
Proof of Burn (PoB)	Validators demonstrate commitment by permanently destroying cryptocurrency (burning), gaining proportional validation rights.
Byzantine Fault Tolerance (BFT)	Achieves consensus through voting-based mechanisms, ensuring the blockchain remains functional even when nodes are faulty or malicious.

than classical computers. Additionally, quantum miners possess a computational advantage in mining compared to their classical counterparts [2, 8, 29]. Shor's algorithm can break many existing cryptographic approaches such as RSA, Diffie-Hellman, and Elliptic Curve. Consequently, consensus protocols that rely on classical cryptographic operations such as digital signatures (e.g. Proof of Stake - PoS) for block-signing and verifications are vulnerable to quantum attacks. Table 2 provides an overview of the impact of Shor's and Grover's algorithms on current pre-quantum protocols.

Post-quantum cryptography and quantum cryptography are known to potentially resist quantum attacks. Collectively, we will refer to both approaches as quantum-resistant (or quantum-safe) in this paper when we review and discuss the extant literature on such consensus protocols.

We will now describe the layout for the remainder of this paper. Section 2 introduces the basics of post-quantum cryptography with a focus on the mathematical aspects of lattice-based and multivariate cryptography, which are two key areas of interest in the reviewed studies. In Section 3 we describe our literature review methodology, prior to discussing existing consensus protocols in Section 4. The last section concludes this work.



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Table 2: Impact of Shor's and Grover's Algorithms on Pre-Quantum Consensus Protocols

Protocol	Cryptography	Shor	Grover	Mining Advantage	Comments
Proof-of-Work (PoW)	Hash functions (SHA-256)	No	Yes (quadratic speedup)	Quantum miners gain $\sqrt{N}$ speedup	Requires longer hashes for future security
Proof-of-Stake (PoS)	RSA, ECDSA	Yes	Limited	No direct mining; block forging at risk	Shor's breaks signature-based validation
BFT	RSA, ECDSA	Yes	Limited	– (no mining)	Signature schemes vulnerable
Secret Leader Election	Digital sigs, threshold crypto	Yes	Limited	– (no mining)	Signatures/threshold crypto vulnerable
Proof of Luck	Trusted HW, sigs	Yes	Limited	– (no mining)	Attestation/signatures breakable
Proof of Burn	Hash, digital sigs	Yes (sigs), No (hash only)	Yes (hash)	Quantum mining advantage or forgery if sigs used	Signature-based burns breakable; hash-based quantum speedup

2 Mathematical Background

Post-quantum Cryptography (PQC) is a field aiming at the design and analysis of cryptographic systems that resist attacks from both classical and quantum computers, and can interoperate with existing communications protocols and networks. Important classes of PQC schemes are Lattice-based, Multivariate-based, Code-based, and Hash-based cryptography. In this section, we focus on the basic mathematical background of Lattice-based and multivariate cryptography and the most common intractable problems upon which they rely. For more in-depth information on these topics, we refer the reader to [4, 11, 15, 40, 41].

2.1 Lattice-based Cryptography

Lattice-based cryptography refers to a wide range of cryptographic primitives whose security is based on mathematical problems related to lattices [4, 28].

A lattice $L(B)$ in $\mathbb{R}^n$ is defined as the set of all integer linear combinations of the rows of a matrix $B \in \mathbb{R}^{m \times n}$ with linearly independent rows, i.e., $L(B) = \{xB \mid x \in \mathbb{Z}^m\}$. The matrix B is called a basis for $L(B)$, while m and n are called the rank and dimension of $L(B)$, respectively. If $m = n$, then $L(B)$ is called a full-rank lattice. For cryptographic applications, we usually consider q -ary lattices. These lattices are full-rank lattices and are defined by a matrix $B \in \mathbb{Z}^{m \times n}$ as follows: $L_q(B) = \{v \in \mathbb{Z}^n : v \equiv xB \pmod{q} \text{ for some } x \in \mathbb{Z}_q^m\}$. Two important concepts in Lattice-based cryptography are the length of a vector in $\mathbb{R}^n$ and the length of a basis of a lattice. The length of a vector $x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ is defined as $\|x\| = (x_1^2 + \dots + x_n^2)^{1/2}$, while the length of a basis B of a lattice is defined as $\ell(B) = \max \{\|s\| : s \text{ is a row of } B\}$.

The security of cryptographic schemes based on lattices relies on certain lattice problems, some of which are believed to belong to class NP-hard and intractable by both classical and quantum computers. We formulate some of these problems next.

Shortest Vector Problem (SVP): Given a lattice $L(B) \subseteq \mathbb{R}^n$, find a shortest non-zero vector in $L(B)$.

Shortest Independent Vector Problem (SIVP): Given a lattice $L(B) \subseteq \mathbb{R}^n$ of rank m , find a basis $\mathcal{B}$ of minimum length among all basis of $L(B)$.

Closest Vector Problem (CVP): Given a lattice $L(B) \subseteq \mathbb{Z}^n$ and a vector $x \in \mathbb{R}^n$ that is not in $L(B)$, find a vector $v \in L(B)$ that is closest to v .

Learning with Errors (LWE): Given n, m, q positive integers with $m \geq n$ and $q \geq 2$. Assume $A \in \mathbb{Z}_q^{m \times n}$ and $s \in \mathbb{Z}_q^n$ are chosen uniformly. Let χ be a probability distribution on $\mathbb{Z}$ and $e \in \mathbb{Z}_q^m$ a vector that is drawn from the distribution χ^m . The problem LWE is to find s given A and $b = As + e$.

Practical lattice-based cryptographic primitives are typically formulated using approximation variants of SVP, SIVP and CVP. The accuracy of the approximation is controlled by a parameter $\gamma \geq 1$, which is computed in function of the lattice. For example, in the approximation variant of SVP, often denoted SVP_γ , the target is to find a non-zero vector $v \in L(B)$ so that $\|v\| \leq \gamma t$, where t denotes the length of a shortest vector in $L(B)$. In the same vein, the LWE problem has been expanded to use algebraic rings; a problem known as Ring-LWE.

To solve the lattice problems mentioned above, lattice basis reduction algorithms such as the Lenstra-lenstra-Lovász (LLL) algorithm, the Block Korkin-Zolotarev (BKZ) algorithm or the Seysen Reduction algorithm are applied. The goal in these algorithms is to reduce the basis B of a lattice $L(B) \subseteq \mathbb{Z}^{m \times n}$ to a basis with shortest vectors that are nearly orthogonal. These lattice reduction algorithms provide only a solution to the approximation variants of the problems mentioned above. For example, the LLL algorithm has a polynomial running time of $O(n^5 m \log(\ell(B))^3)$, but has an approximation factor of order $2^{O(n)}$, where n is the dimension of the lattice. On the other hand, the BKZ algorithm has a running time of $2^{O(\beta \log(\beta))}$ with an approximation factor of $\beta^{O(n/\beta)}$, where β is an algorithm parameter chosen from the set $\{2, 3, \dots, n\}$. Other known algorithms that aim to find an exact solution or even an approximation with polynomial factors, have running times

either $2^{O(n)}$ or $2^{O(n \log(n))}$, but require an exponential or large polynomial memory space, which makes them essentially impractical (c.f. [4]). It is then conjectured that lattice-based cryptographic primitives are secure even when attacked by quantum computers. In addition to their conjectured security, the advantages of some lattice-based cryptosystems are the relative ease of implementation, small keys as well as the potential to adapt them for other purposes, like cryptosystems providing fully homomorphic encryption.

2.2 Multivariate Cryptography

Multivariate cryptography refers to a class of cryptographic schemes that use multivariate polynomials over a finite field in combination with linear and non-linear mappings. This section aims to give an overview of multivariate cryptography. For additional information on multivariate cryptography, the reader is referred to [4, 11].

Let $\mathbb{F}_q$ be the finite field with q elements and $\mathbb{F}_q[x_1, \dots, x_n]$ be the set of all multivariate polynomials with coefficients in $\mathbb{F}_q$. A quadratic polynomial in $\mathbb{F}_q[x_1, \dots, x_n]$ is one having total degree 2. The security of multivariate cryptographic schemes is based on the Multivariate Quadratic problem, which we describe next.

Multivariate Quadratic (MQ) problem: Given integers m, n with $m \leq n$, a finite field $\mathbb{F}_q$, and m quadratic polynomials $f_1, f_2, \dots, f_m$ in $\mathbb{F}_q[x_1, \dots, x_n]$, the MQ problem consists on finding a solution in $\mathbb{F}_q^n$ to the system of quadratic equations:

$$f_1(x_1, \dots, x_n) = f_2(x_1, \dots, x_n) = \dots = f_m(x_1, \dots, x_n) = 0.$$

The MQ problem belongs to the class of NP-complete problems, and using quantum computers does not provide an advantage to solve it. Very important parameters within this problem are m and n . The most difficult instances are obtained when m and n have similar values. If the difference in value between them is very large, some algorithms can efficiently solve the problem.

For cryptographic purposes, the MQ problem is used alongside with the Isomorphism of Polynomials problem.

Isomorphism of Polynomials (IP) Problem: Given two functions $A : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$ and $B : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$ specified by m -tuples of polynomials over $\mathbb{F}_q$, find two affine transformations S and T such that $A = S \circ B \circ T$. The affine maps S and T mask the structure of the polynomials defining the function B .

Currently, the most effective algorithms for solving the MQ problem are based on techniques that utilize Gröbner basis, which can be understood as an extension of a Gaussian elimination to systems of multivariate polynomials. The computational cost of computing Gröbner basis increases rapidly with the size of the problem. This makes it impractical for an appropriate selection of the parameters m, n, q and the polynomials $f_1, \dots, f_m$. Another algorithm for solving the MQ problem is the eXtended Linearization algorithm which intuitively transforms a quadratic multivariate system into a linear system of equations with a larger number of variables.

Multivariate cryptosystems seem to be a good choice mainly for the construction of digital signature schemes; see [11]. In this direction, some digital signature multivariate schemes provide very short signatures as compared to other post-quantum signature schemes like lattice-based and hash-based schemes. Moreover, the actual computational operations require additions and multiplications of

small field elements that are fast and simple to implement even on constrained hardware. On the other hand, the largest issue of multivariate cryptographic schemes is the size of the public keys. They are large when compared to other schemes. In contrast, private keys are smaller, but still formidable for small embedded devices which have memory constraints.

3 Search Methodology

We conducted a systematic search for relevant articles on Google Scholar using keywords and Boolean operators, including “quantum” AND “post-quantum” AND “consensus protocols”. The initial search yielded several hundred papers. However, upon closer examination, many were found to be irrelevant and were subsequently excluded. In the second screening phase, 176 papers were selected for further evaluation based on their abstracts and content. After a thorough review, 25 articles were deemed relevant and included in our study. These articles specifically present new quantum-resistant consensus protocols for blockchain, encompassing quantum, post-quantum, and information-theoretic approaches. Figure 1 graphically describes our search methodology.

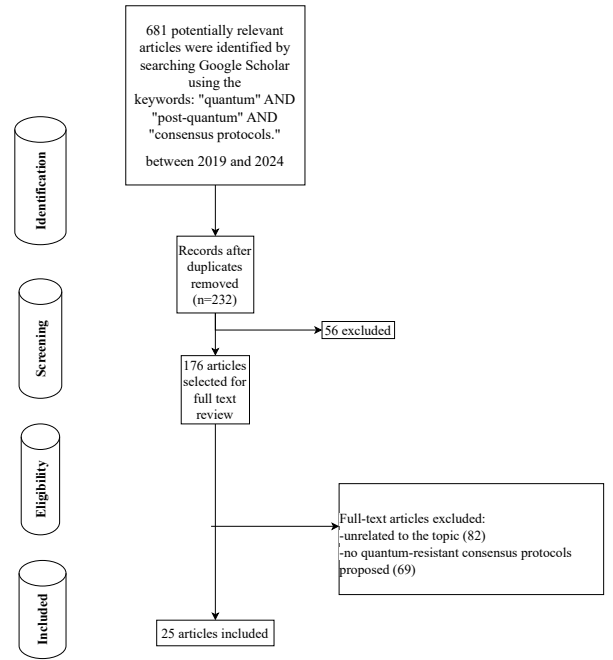


Figure 1: Literature Search Protocol

We also identified a small number of similar review articles [19, 20, 22]. Jose and Panchami [22] reviewed eight post-quantum consensus protocols published between 2017 and 2022, and these protocols were categorized based on problem type (e.g., NP-hard problems). However, this study excludes quantum protocols and does not comprehensively evaluate the security or performance of the proposed protocols. Gomes et al. [20] systematically reviewed 29 articles published between 2018 and 2022. While the review

Table 3: Similar Literature Reviews: A Comparative Summary

Study	Number of Articles Reviewed	Scope	Analysis Focus
Jose, Jisha Mary, & Panchami (2022) [22]	8 between 2017 and 2022	Post-Quantum consensus protocols	Classification of post-quantum protocols
Gomes, Khan, and Voinovich (2023) [20]	29 between 2018 and 2022	Quantum and Post-Quantum consensus protocols	Classification of quantum-resistant protocols with emphasis on performance and adoption barriers
Fernandez-Caramels and Fragua-Lamas (2020) [19]	Not specified, between 2000 and 2020	Post-Quantum cryptography and its applicability to blockchain	Classification of post-quantum protocols and their applicability to blockchain
Our Study	25 between 2019 and 2024	Quantum and Post-Quantum consensus protocols	Foundational classification of quantum-resistant protocols with emphasis on security properties

covers both quantum and post-quantum consensus protocols, the focus is primarily on classifying post-quantum consensus protocols and their performance and adoption barriers. However, security properties and evaluations were not comprehensively studied. Fernandez-Caramels and Fraga-Lamas [19] did not specifically focus on consensus protocols, as they focused on the threat posed by quantum computing to blockchain. They also explored various types of post-quantum cryptosystems, and their applicability to blockchain systems. This study provides detailed comparisons of different post-quantum cryptosystems, including code-based, lattice-based, and multivariate-based schemes, along with their performance in blockchain applications. However, the study offers limited insights into consensus protocols and their quantum resistance. Additionally, there is no explicit mention of the number of articles reviewed or the publication years; though we observed that the included papers date from 2000 to 2020. Table 3 comparatively summarizes existing survey papers [19, 20, 22] and our work.

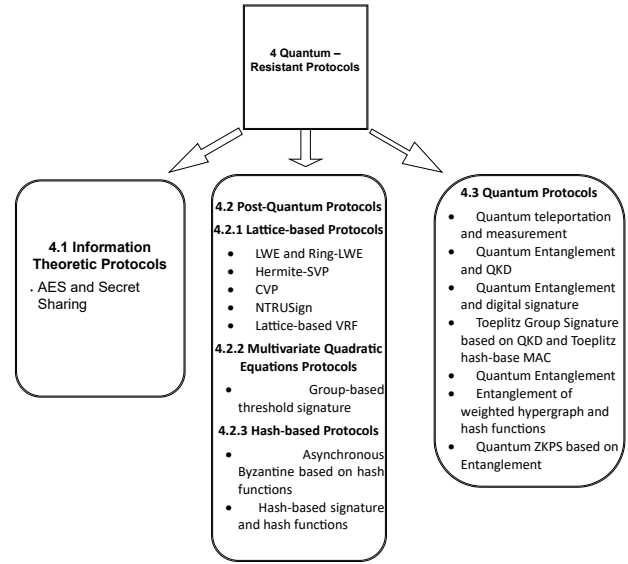
4 Existing Consensus Protocols

Quantum-resistant blockchain consensus protocols can be broadly categorized into three types according to their security properties and assumptions. The first category is based on information-theoretic security, which relies on randomness and perfect secrecy. The second category is based on PQC primitives (such as those described in Section 2). The third category consists of quantum cryptographic protocols which leverage the principles of quantum mechanics such as superposition and entanglement for security. Thus, we found it beneficial to group these protocols based on their security properties and assumptions, providing a foundational classification to compare their resistance to quantum threats. Table 6 provides an overview of the characteristics and limitations of these categories.

4.1 Information Theoretic Protocols

Protocols in this category rely on randomness and perfect secrecy to provide unconditional security, which does not depend on computational limits. Cryptographic systems, such as the One-Time Pad, use random keys to achieve perfect secrecy, making them immune to quantum attacks.

Dolev et al. [12], Dolev and Wang [13, 14] proposed two asynchronous Byzantine fault-tolerance (aBFT) protocols (i.e., SodsBC and SodsBC++), designed to achieve quantum resistance by using information-theoretic secure cryptographic components. The authors also proposed a framework that resists quantum adversaries

**Figure 2: Taxonomy of Protocols Based On Security Properties**

while also meeting the performance needs of modern blockchain applications using symmetric encryption, namely: AES, and perfectly secures secret-sharing schemes establishing security independently of computational assumptions. Both protocols are designed to operate in an asynchronous network where no timing assumptions are required for message delivery. This ensures that the system can handle arbitrary network delays without compromising correctness. There are two consensus mechanisms employed:

- BBA-ACS (Binary Byzantine Agreement - Asynchronous Common Subset): Used in Sods. This architecture includes multiple parallel BBA instances, making it suitable for relatively smaller networks.
- MVBA-ACS (Multi-Value Byzantine Agreement - Asynchronous Common Subset): This architecture is employed in SodsBC++ and uses a single MVBA instance to improve efficiency, especially in larger networks.

The consensus process begins with each participant proposing encrypted transactions, sharing cryptographic keys and random values through an integrated broadcast, and secret-sharing protocol.

Table 4: Summary of Consensus Mechanisms and Security Properties

Study	Consensus Mechanism	Security Properties/Assumption	Security Proof Type	Main Problem Addressed
Chen et al. [6, 7]	Post-quantum PoW	Hardness of the Multivariate Quadratic (MQ) problem	Not Given	Mining Advantage & Cryptographic operations
Ding [10]	Post-quantum PoW	Hardness of the Multivariate Quadratic (MQ) problem	Not Given	Mining Advantage
Mi et al. [27]	Post-quantum PoW	Hardness of the approximate CVP	Not Given	Cryptographic operations
Boneh et al. [5]	Post-quantum Secret Leader Election	Hardness of LWE and Ring LWE	Game-Based proof	Cryptographic operations
Yi et al. [39]	Post-Quantum Threshold Signature-based Consensus	Hardness of solving quadratic equations in finite fields	Not Given	Cryptographic operations
Das et al. [9]	Asynchronous Common Subset (ACS) Byzantine consensus	Security of Hash Functions	Random Oracle Model	Mining Advantage & Cryptographic operations
Endurthi et al. [16]	Post-quantum PoW	Hardness of CVP	Not Given	Mining Advantage
Far et al. [18]	Post-quantum Proof of Burn	Security of SHA-3	Not Given	Cryptographic operations
Wang et al. [36]	An improved version of PoW with serial mining puzzle	Security of Cryptographic Hashing	Not Given	Mining Advantage
Behnia et al. [3]	Post-quantum PoW	Hardness of Hermite Shortest Vector Problem	Complexity proof of efficiency	Mining Advantage
Kim et al. [24] [23]	Delegated Proof of Luck	Lattice based Verifiable random functions & Falcon-512 (Lattice based signature)	Not Given	Cryptographic operations
Esgin et al. [17]	Lattice-Based Verifiable Random Function	Module-LWE, uniqueness under Module-SIS	Formal cryptographic reduction	Cryptographic operations
Peng et al. [30]	Consensus-based editor election using time capsules and chameleon hash collisions	Lattice hardness and hashing	Not Given	Cryptographic Operations
Dolev and Wang [13] [14] [12]	Asynchronous BFT	Perfect information-theoretic secrecy schemes and nested hashing	Not Given	Cryptographic operations
Wen et al. [38]	Consensus based on quantum zero-knowledge proof	Quantum Teleportation and measurement	Not Given	Cryptographic operations
Wen et al. [37]	Consensus Mechanism Based on Quantum Teleportation	Quantum Teleportation and measurement	Not Given	Cryptographic operations
Wang and Yu [35]	Consensus Protocol Based on Quantum Attack Algorithm	Irreversibility, and uncertainty of quantum measurement	Not Given	Cryptographic operations
Li et al. [25]	Quantum delegated proof of stake	Quantum voting and quantum digital signature	Theoretical Security Proof	Cryptographic operations
Sun et al. [34]	Quantum Based BFT	Quantum-secured Toeplitz Group Signatures and QKD	Theoretical Security Proof	Cryptographic operations
Banerjee et al. (2020) [1]	Consensus Using Weighted Hypergraph States	Entanglement in quantum weighted hypergraph states	Not Given	Cryptographic operations
Luo et al. [26]	Multi-Valued Detectable Byzantine Agreement for multi-valued inputs	Quantum Entangled states	Not Given	Cryptographic operations

Participants then reach consensus on the proposed transactions through parallel BBAs in SodsBC or a single MVBA in SodsBC++. After consensus, participants reconstruct shared AES keys to decrypt the transactions, forming the final block.

The utilization of secret sharing in this protocol ensures that encrypted data maintains perfect secrecy as long as fewer than the required number of shares are compromised. The post-quantum random coin generation is based on collaborative secret sharing, which ensures that no participant can manipulate the randomness or predict the outcome. The use of AES with an adequate key length increases resistance against quantum attacks using Shor's algorithm. Although these techniques are promising, the authors do not provide a formal security analysis and proof to demonstrate the robustness of their proposed protocol.

4.2 Post-Quantum Protocols

Protocols in this category rely on the computational difficulty of solving certain mathematical problems. Although code-based cryptography is a promising solution to resist quantum attacks, we have not yet found a consensus algorithm that utilizes code-based cryptographic primitives for blockchain consensus.

4.2.1 Lattice Based Protocols. Boneh et al. [5] proposed a consensus protocol referred to as Single Secret Leader Election (SSLE). The latter is designed to enable a group of participants to elect a single leader secretly, meaning that only the elected leader knows their identity until they choose to reveal it. The primary goal of the protocol is to enhance the security of consensus mechanisms in blockchain, particularly in Proof of Stake (PoS) systems, by keeping the leader's identity hidden to prevent potential denial-of-service attacks or other targeted actions. The protocol operates on two main phases, registration and election. During the registration phase, participants register by publishing public information (commitments) on a bulletin board while keeping associated secret information private. This involves generating a commitment based on a secret key and a re-randomizable commitment scheme, which is based on the lattice problems Learning With Errors (LWE) and its variant Ring-LWE. During the election, participants use a random beacon along with public commitments to choose a leader randomly. The chosen leader can later reveal themselves by providing proof of their selection through their secret key. This process involves "commit-and-shuffle," where commitments are re-randomized and shuffled by the participants, ensuring unpredictability and unlinkability of the leader's identity.

The protocol achieves security through binding, fairness, and unpredictability, where binding ensures that a commitment can only be linked to one leader, so only the correct leader can reveal themselves with proof, fairness ensures that all parties have an equal probability of being chosen, and malicious parties cannot influence the outcome, and unpredictability ensures that the leader's identity remains hidden from all parties until they reveal themselves. The paper uses game-based proofs to model security properties, while oracle-based techniques demonstrate the protocol's resilience to attacks, assuming the random oracle model to hash commitment values.

Employing a secure and efficient signature scheme is a critical consideration in post-quantum consensus protocols due to the inherent computational complexity of such schemes. To address that, Kim et al. [24] introduced the Delegated Proof of Luck consensus protocol. The latter incorporates the Dilithium signature scheme, one of the signature algorithms selected by NIST for post-quantum cryptographic applications. Dilithium utilizes SVP and is employed by Kim et al. [23, 24] in their Proof of Luck mechanism with a delegated approach to enhance block generation speed while maintaining post-quantum security. Proof of Luck uses randomness to select block producers while also including Trusted Execution Environment (TEE) which enforces fair block generation and voting processes to prevent cheating.

In the delegation phase, nodes generate random numbers as votes, broadcast them with Dilithium signatures, and verify these transactions. The top nodes with the most votes are selected as delegates to participate in the consensus process, reducing the overall computational load. In the consensus phase, delegated nodes generate and verify blocks using Trusted Execution Environment (TEE) mechanisms. This involves: (1) Generating random "luck" values and proof through TEE; (2) Validating block components like nonce, proof, and previous hash; and (3) Verifying transactions within blocks using Dilithium signatures. The potential advantage of this approach lies in the combination of TEE-based verification and the delegation process, which improves the overall efficiency of the consensus mechanism.

Behnia et al. [3] introduced a novel Proof of Work that leverages the Hermite Shortest Vector Problem (Hermite-SVP) in lattice-based cryptography. The proposed mechanism aims to reduce the computational advantage of quantum computers over classical ones. In this protocol, miners attempt to solve an instance of Hermite-SVP by finding a lattice vector v that satisfies the condition: $\|v\| \leq \alpha \cdot p^{1/n}$ where $p = \det(\Lambda)$ is the lattice determinant, and α is an approximation factor determining how close the found vector is to the shortest vector.

Solving this problem, which involves finding a lattice vector of minimal length, is known to be NP-hard, and thus is believed to resist both classical and quantum attacks. The difficulty of the puzzle can be fine-tuned by adjusting the lattice dimension n , where higher dimensions increase the computational cost of solving the SVP problem, making it scalable with the growth of computational resources. Challenges are generated with randomness derived from the previous block to ensure that pre-computation attacks are infeasible.

Behnia et al. [3] showed that quantum computers gain only a marginal asymptotic advantage in solving Hermite-SVP compared to classical computers. The best known quantum algorithm achieves complexity $2^{0.265n+o(n)}$ compared to $2^{0.292n+o(n)}$ for classical algorithms. Although their proposed protocol aims to maintain fairness between classical and quantum miners, its practicality, however, will depend on advancements in efficient lattice-based computations and broader adoption of post-quantum cryptographic frameworks. Additionally, the protocol does not address quantum attacks on other layers of blockchain security, such as cryptographic hash functions or digital signatures.

Similarly, Endurthi et al. [16] reconstructed the consensus mechanism proposed by Behnia et al. [3] based on CVP instead of SVP. Since CVP is considered more challenging to approximate than SVP, it offers greater security against both classical and quantum attacks. However, the paper carries forward the limitations of its predecessor and does not provide insight on practical implementation.

Another critical component in protecting blockchain consensus is digital signatures, because they ensure the authenticity, integrity, and non-repudiation of transactions. Mi et al. [27] introduced a PoW scheme that replaces traditional hash-based puzzles with a probabilistic signature mechanism based on NTRUSign, a lattice-based signature scheme. The proposed PoW mechanism uses NTRUSign as the basis for block generation, with the signing process itself serving as the proof of work. Instead of requiring participants to solve hash puzzles, they must generate valid signatures to earn the right to add blocks to the chain. The difficulty of this protocol is adjusted by modifying the probability of successfully generating a valid NTRUSign signature, making it possible to control the success rate of miners without relying on hash-based difficulty adjustments. The miner's signing keys are tied to their block creation process, reducing the likelihood of unauthorized tampering or misuse. The success rate for generating valid signatures is governed by a probabilistic model using Burr distributions within a gamma distribution family. This probability can be fine-tuned by adjusting parameters in the signing process to control the workload expected of miners.

The scheme proposed Mi et al. [27] is based on approximate closest vector problems (appr-CVP). Although the original NTRU-Sign scheme was broken due to lattice-based attacks and key leakage, the modified versions mitigate these vulnerabilities by increasing the key size, improving randomness, and adopting techniques like rejection sampling. These improvements make it a stronger candidate for post-quantum digital signatures.

Esgin et al. [17] introduced Lattice-Based Verifiable Random Function (LB-VRF) based on lattice problems, specifically Module-SIS and Module-LWE). The goal is to replace quantum-vulnerable VRFs, with a quantum-secure alternative for blockchain protocols to prevent possible "history rewriting" attacks in the future. The LB-VRF is tested with Algorand's consensus protocol, which is designed to support PoS systems where committee members are selected randomly and verifiably. The LB-VRF ensures that the selection of committee members cannot be manipulated, preserving the integrity of the consensus process. The consensus protocol begins with each node generating a lattice-based public-private key pair for the VRF.

At the start of each round, the nodes use their VRF to compute a random value and a proof based on the input of the round and their private key. The node with the lowest VRF output is probabilistically selected as the leader for that round, and the accompanying proof ensures the fairness and correctness of the selection process. The leader then proposes a new block containing pending transactions, along with the VRF output and proof to validate its eligibility. Other nodes verify both the VRF proof and the validity of the proposed block before participating in the final consensus mechanism to agree on the block. Once verified and accepted, the new block is added to the blockchain, and the process is repeated for the next round. This approach ensures fair, efficient, and post-quantum

secure consensus, leveraging VRF for random leader selection and lightweight verification.

The construction uses the "Fiat-Shamir with Aborts" technique, which allows for zero-knowledge proofs of knowledge of the secret without excessive computational overhead. Uniqueness and pseudorandomness properties are derived from lattice-based problems, but the design restricts VRF key usage to a few outputs per key to maintain security. Esgin et al. [17] included a proof of security for LB-VRF, demonstrating pseudorandomness under the MLWE assumption and computational full uniqueness under the MSIS assumption. The uniqueness proof is achieved through a "double rewinding" argument, ensuring that the VRF outputs remain unique for a given input and public key under standard post-quantum assumptions.

4.2.2 Multivariate Quadratic Equations Based Protocols. Yi et al. [39] proposed a novel blockchain consensus algorithm based on a post-quantum threshold signature scheme to address the inefficiencies of existing consensus mechanisms such as PoW and PBFT. Using this signature scheme, the authors aim to reduce the energy consumption and communication overhead that are typically associated with other consensus protocols, while ensuring scalability, decentralization, and security against quantum attacks.

The proposed protocol uses a group-based threshold signature scheme to achieve consensus on new blocks with the following key components: (1) Random Group and Leader Selection: The network is divided into groups. Each group randomly selects a leader to propose a new block. (2) Threshold Signatures for Block Validation: Once the leader generates a block, the group members participate in the process of block validation through a threshold of nodes such that more than half must sign the block. (3) Manager Nodes: Manager nodes oversee the operations of the group and ensure that the processes of leader selection and block validation run smoothly. The protocol aims to reduce the energy demands seen in PoW systems and minimize the communication overhead associated with PBFT. This threshold signature that the protocol employs reduces the number of required communications for consensus, which allows it to scale to large networks effectively.

The protocol proposed by Yi et al. [39] is grounded in the use of post-quantum threshold signatures, which is based on the hardness of the MQ problem. However, there is no formal security proof provided.

Chen et al. [6] introduced a post-quantum PoW consensus protocol and an Identity-based post-quantum signature scheme. The PoW protocol uses the MQ problem as the core of the mining challenge. The identity-based signature scheme, referred to as ID-Rainbow, is used to secure block chain transactions and reduce key size, which also provides additional contribution to its security robustness. The proposed mechanism is built around a post-quantum PoW model that integrates both computational and memory mining capabilities. This improves the security and efficiency of traditional blockchain models, such as Bitcoin. The key components are as follows:

(1) PoW

- The core of the proposed PoW mining challenge involves solving a system of multivariate quadratic equations. The complexity of solving these equations makes the system

quantum resistant, unlike Bitcoin's SHA-256, which is vulnerable to quantum attacks.

- The protocol introduces a dynamic difficulty adjustment algorithm (DAA) that adapts the difficulty based on network conditions to ensure a stable rate for block generation.
- (2) Identity-Based Quantum Resistant Signature Scheme
- The ID-Rainbow scheme reduces the size of public keys and signatures to 8 bytes and 46 bytes, respectively, which improves both security and efficiency.
 - The scheme supports lightweight transactions by embedding the identity-based public key directly into the transaction structure, reducing the computational overhead compared to traditional signature schemes such as Elliptic Curve Digital Signature Algorithm (ECDSA).
- (3) Efficient Blockchain Architecture
- The blockchain architecture includes *miner nodes*, *winning nodes*, and *common nodes*.
 - Miner nodes solve the quadratic equations using the Gröbner basis algorithm.
 - Winning nodes broadcast the solution to the network and validate the block.
 - The protocol enables memory mining, which requires miners to use both computational and storage resources to mine blocks.

The authors employed the same protocol for organizing various devices in a smart city in [7]. However, no formal proof is presented to demonstrate the security of the protocol. Similarly, Ding [10] proposed a PoW mechanism that relies on the hardness of the MQ problem. The authors aimed to show that the protocol achieves certain properties such as intrinsic hardness, public verifiability, and independent distribution. However, several of these properties are not rigorously discussed. For example, intrinsic and homogenous hardness are assumed based on the fact that random MQ problems are NP-Hard. However, there is no discussion of the worst-to-average case reduction connection, an important consideration as we will discuss in the later section.

4.2.3 Hash-based Protocols. Das et al. [9] introduced a novel asynchronous Byzantine consensus mechanism that achieves agreement among distributed parties without relying on public-key cryptography. Instead, it uses cryptographic hash functions modeled as a random oracle to ensure post-quantum security.

- (1) Parties start by sharing secrets securely using hash-based commitments (ASKS).
- (2) Honest parties validate each other's contributions and agree on a set of indices (ICG).
- (3) A weak leader is probabilistically elected based on random ranks.
- (4) The protocol iteratively resolves conflicts and ensures that all honest parties agree on the same leader's output (ACS).

The security of this protocol is based on the assumption that hash functions modeled as a random oracle are collision-resistant. By eliminating the need for public-key cryptography, such as digital signatures or trusted setups, the protocol enhances post-quantum security, as it does not rely on structures vulnerable to quantum

attacks. The protocol uses a weak common coin that provides probabilistic agreement among honest nodes without requiring strong threshold signatures. This coin generation process is integral for leader election within the protocol's framework. The Index Cover Gather (ICG), which prevents adversarial manipulation of leader election, ensuring that the selected leader is agreed upon by honest parties and minimizing the impact of malicious actors. Using the Random Oracle Model (ROM), the authors present a proof that the adversary's ability to manipulate the protocol is limited to querying the oracle and observing the results, formalizing the unpredictability of the hash function. Furthermore, the authors adopt a game-based approach to define and prove the protocol's security properties such as agreement, termination, and validity. The adversary's advantage of winning these games is formally analyzed and shown to be negligible under the given assumptions.

Another approach that aims to provide a comprehensive framework to transition existing cryptocurrencies, such as Bitcoin, to quantum-secure versions is proposed by Far et al. [18]. The framework proposes using a hard fork to create a quantum-secure version of Bitcoin, named Quantum Bitcoin (QBTC), utilizing hash-based signatures and quantum-secure hash functions. The framework suggests a Proof of Burn (PoB) mechanism for transitioning assets to the new quantum-secure cryptocurrency. In this mechanism, Users "burn" their existing coins by sending them to an unspendable address, proving ownership in a 1-to-1 conversion to obtain the new quantum-secure coins (QBTC). The mining process remains similar to traditional PoW, but SHA-3 is used instead of SHA-256, and hash-based signatures replace ECDSA for transaction signing, adding quantum resistance. The security of the framework depends on quantum resistant primitives, including hash functions (SHA-3) and hash-based signatures, such as Merkle signatures. Although these primitives provide security against Shor's algorithm, the paper does not offer a detailed cryptographic proof, but relies on the assumption that quantum-secure hash functions and signatures are sufficient for post-quantum security.

Another approach to mitigate the quantum advantage of both parallel and quantum mining, Wang et al. [36] introduced an enhanced PoW mechanism designed to address resource centralization and provide resistance to quantum attacks by leveraging hashing as its core cryptographic primitive. The protocol achieves post-quantum security by enforcing serial computation through a Serial Mining Puzzle (SMP) and improving fairness with a Mining Credibility System (MCS). The SMP ensures miners solve puzzles sequentially by constructing a hash chain, where each step of the puzzle depends on the solution of the previous step. Unlike traditional PoW puzzles, which benefit from parallel computation, SMP can only be solved in a serial manner, minimizing the advantage of using multiple CPUs or specialized hardware as well as reducing the potential computational advantage of quantum computers. The reliance on cryptographic hash functions prevents miners from parallelizing computations, reducing the quantum advantage provided by Grover's algorithm. The MCS evaluates miners' credibility based on their past behavior. Miners with higher credibility enjoy lower mining difficulty, which discourages dishonest behavior. Credibility is updated over time based on various mining events, like creating valid or invalid blocks, or verifying blocks. This process is adjusted according to each miner's credibility, with honest

miners being rewarded with easier mining tasks, while dishonest miners face increased difficulty, which makes cheating costly. The protocol maintains key blockchain security guarantees, such as common prefix, chain quality, and chain growth, while being resistant to quantum adversaries. However, no formal security proof is provided in the paper.

Peng et al. [30] proposed a post-quantum redactable blockchain scheme that allows modifications to historical blockchain data while maintaining its integrity and security. The protocol relies on lattice-based chameleon hash functions, which is a cryptographic primitives used to ensure malleability (the ability to modify a message's hash without revealing the modification). These hash functions allow trapdoor-based modification which ensures that only legitimate editors with the secret key can compute hash collisions. The objectives of the design as to approve edits to historical blocks in a blockchain through multi-party consensus, elect a block editor fairly to perform the edits using the lattice-based chameleon hash function, and maintain the integrity and trust of the blockchain, ensuring no malicious edits are made without proper agreement.

The protocol proposed by Peng et al. [30] enables controlled modifications to blockchain data through a multi-step process. First, a node proposes an edit to a historical block, providing justification and metadata to the network. The proposal is then subjected to a voting process, where nodes collectively decide whether to approve the edit. If approved, a random editor is elected using a time capsule-based mechanism to ensure fairness and unpredictability. The selected editor performs the modification using the lattice-based chameleon hash function, which allows controlled hash collisions while keeping the hash value unchanged with a secret trapdoor. The modified block, along with the updated chameleon hash, is transmitted to the network for verification. All nodes verify the correctness of the edit, ensuring that the chameleon hash is satisfied and the editor's actions are valid. Finally, the network reaches consensus on the modified blockchain, ensuring transparency, integrity, and post-quantum security throughout the process. The security of the scheme is based on the hardness of the SIS problem and the collision resistance chameleon hash function, while the randomized editor election prevents adversarial manipulation.

4.3 Quantum Protocols

Protocols in this category utilize principles of quantum mechanics, such as quantum teleportation and entanglement, to facilitate consensus in distributed systems. By leveraging quantum communication methods, these protocols aim to achieve secure and efficient agreement among network participants. Wen et al. [37] proposed a novel quantum blockchain consensus mechanism based on quantum teleportation and quantum measurement. This design aims to improve the classical consensus mechanisms by offering unconditional security, energy efficiency, and high throughput while protecting against 51% attacks. First, a quantum channel is initialized using entangled states (EPR pairs), ensuring secure communication between users and miners. Users generate a random nonce S through quantum measurement, guaranteeing inherent unpredictability. The miners then generate a candidate nonce S' , and quantum teleportation is used to securely transfer S' to the user for comparison without revealing S to the miner. The verification step

Table 5: Characteristics and Limitations of Cryptographic Categories.

Category	Characteristics	Limitations
Information-theoretic protocols	Security does not depend on computational assumptions; remains secure against any adversary. Examples: one-time pad, secret sharing.	Impractical key distribution for large-scale use; requires pre-shared secrets; limited flexibility.
Post-quantum protocols	Based on problems believed hard for quantum computers; usable with current infrastructure; includes code-based, lattice-based, etc.	Security relies on unproven assumptions; some have large key sizes; real-world performance varies.
Quantum protocols (e.g., QKD)	Rely on quantum mechanics for security; can provide provable secrecy in key distribution.	Require specialized hardware; limited by distance and noise; difficult integration with classical systems.

involves quantum measurements to check if S' matches the original nonce S . If the match is successful, the miner is granted the right to create a new block, which is broadcast to the network for validation. Nodes verify the block and ensure it adheres to the protocol before appending it to the blockchain. By eliminating computationally intensive PoW and relying on quantum mechanics, the protocol achieves fairness, decentralization, and quantum resistance while significantly improving energy efficiency.

Wen et al. [37] stated that their proposed mechanism achieves the following properties:

- **Unconditional Security:** The security of the consensus mechanism is derived from the physical principles of quantum mechanics rather than computational hardness assumptions, making it theoretically immune to attacks regardless of the adversary's computational resources. The security is claimed to be unconditional, as it is based on quantum measurement and the no-cloning theorem of quantum mechanics, which ensures that an adversary cannot duplicate quantum states.
- **Zero-Knowledge Property:** The quantum teleportation process ensures that miners cannot deduce any information about the user's nonce during verification, maintaining privacy, and preventing unauthorized access to transaction data.
- **Resistance to 51% Attack:** Since the mechanism does not rely on computational power for consensus, it inherently resists 51% attacks, which are common in classical blockchain systems reliant on computational resources.

Similarly, Wang and Yu [35] proposed a blockchain consensus protocol that leverages quantum measurement principles, including stochasticity, irreversibility, and uncertainty. Quantum entanglement is utilized to facilitate a distributed consensus similar to Byzantine Fault Tolerance (BFT). The proposed consensus protocol leverages quantum measurement and quantum key distribution (QKD) to achieve secure and efficient agreement among blockchain nodes. First, nodes securely communicate using quantum keys generated through QKD, ensuring confidentiality and preventing

tampering. During the consensus process, each node votes on the proposed block using quantum states— $|0\rangle$ for "yes" and $|1\rangle$ for "no"—to represent their decision. A quantum AND operation aggregates the votes, and the majority result is determined through quantum measurements. The quantum properties of stochasticity and the no-cloning theorem ensure that the votes remain secure and cannot be forged or manipulated. Once consensus is reached, the result is broadcast to all nodes for verification, ensuring fault tolerance and trustworthiness. By replacing traditional cryptographic computations with quantum principles, the protocol potentially minimizes energy consumption, achieves unconditional security, and resists quantum attacks. However, the protocol's implementation assumes a quantum-capable environment, which is not yet widely available.

Li et al. [25] proposed Quantum Delegated Proof of Stake (QD-PoS) which is a quantum-based variant of the traditional Delegated Proof of Stake. The consensus mechanism begins with a quantum voting process, where nodes cast votes using entangled quantum states to elect a subset of representative nodes responsible for block generation. The elected nodes take turns producing blocks in a cyclic manner, ensuring efficient and orderly block creation. Each block contains verified transactions and is secured using quantum digital signatures to ensure authenticity and resistance to tampering, even against quantum adversaries. Blocks are then entangled quantumly using weighted graph or hypergraph states, replacing classical hash-based links to provide tamper resistance through quantum mechanics. Misbehaving nodes are penalized, and the representatives are periodically re-elected using quantum voting to maintain fairness and decentralization. By combining quantum entanglement, digital signatures, and voting, QDPoS ensures secure, efficient, and quantum-resistant consensus.

In the protocol proposed by Li et al. [25], the quantum digital signature plays a key role to ensure that no eavesdropping can occur. First, quantum keys are securely distributed using QKD. To sign a transaction, the sender encodes the transaction data into a quantum state, which inherently prevents tampering or forgery due to quantum properties like the no-cloning theorem. For verification, the receiver measures the quantum state using a corresponding measurement basis tied to the sender's key. If the quantum state matches the expected value, the signature is validated. The described signature inherently resists quantum attacks, as adversaries cannot duplicate or manipulate quantum states without detection.

In another independent work, Sun et al. [34] proposed a consensus protocol that uses Toeplitz Group Signature (TGS) that leverages QKD and a Toeplitz hash-based Message Authentication Code (MAC). In the TGS scheme, the QKD protocol is used to securely distribute symmetric keys to all peers in the blockchain network. Each node generates a Toeplitz hash-based MAC using its shared symmetric key and applies it to the message (e.g., transaction data, block proposals). The MAC serves as the group signature, ensuring that the message is authenticated and tamper-proof. Nodes verify the MAC by applying the same Toeplitz hash function with their symmetric keys. If the MAC is valid, it confirms that the message originated from a trusted peer and has not been altered.

In the consensus mechanism, a designated proposing peer creates a block proposal after validating the transactions, signs it with TGS, and sends it to the voting peers. Voting peers validate the proposal, form their own verified blocks, and sign the block hash

using TGS. Votes are exchanged and ordered among peers. Votes are aggregated in the order specified. If a supermajority ($>75\%$) supports a block, it is added to the blockchain. Otherwise, the process continues with the next peer in the order. The protocol ensures unforgeability, transferability, and non-repudiation through TGS. The protocol is designed to tolerate Byzantine faults, allowing the network to reach consensus even if some nodes are compromised or act maliciously. In a quantum-secure network, each node's quantum vote is unique and cannot be copied or altered due to the no-cloning theorem. However, the protocol relies on a limited set of representative nodes to make consensus decisions, following a permissioned blockchain model. Although this setup improves efficiency, it introduces potential centralization risks, as only selected nodes participate in block production and voting, which could lead to power concentration and collusion risks if the representative nodes are controlled by a small group or have overlapping interests.

Another approach to solve the Byzantine agreement problem is offered by Luo et al. [26] who proposed a Quantum Multi-valued Byzantine Agreement (QMBA) protocol based on d -dimensional entangled states. The consensus protocol begins with a trusted party (TP) distributing maximally entangled quantum states to all participants, embedding decoy particles to detect any tampering or eavesdropping during distribution. Each participant measures their quantum particles to generate unique but correlated lists of values. The sender general initiates the process by broadcasting a message and referencing specific positions in their list that correspond to the message. Other generals validate the sender's message by comparing it against their own lists derived from the shared quantum states. If the majority of generals find consistency in the message and the list positions, the message is accepted; otherwise, it is rejected. The use of quantum entanglement ensures tamper-evident communication and the integrity of the process, while classical authenticated channels facilitate the agreement among participants. This quantum-enabled process achieves detectable Byzantine agreement (DBA), ensuring consensus even in the presence of potential adversaries or traitorous participants.

The protocol proposed by Luo et al. [26] extends traditional binary Byzantine agreement to a multi-valued context, making it applicable to more complex decision-making scenarios and aligning with modern blockchain needs. The security analysis of the protocol shows that since the TP is assumed semi-honest and uses decoy states, the distribution process is secure, as generals cannot infer the lists of others beyond what they already know, which maintains confidentiality within the distributed system. Moreover, the protocol guarantees that all loyal generals reach the same decision if the sender is loyal, satisfying the Byzantine agreement conditions. Even if the sender is disloyal, the protocol ensures that all loyal generals agree on a value or abort, thus achieving detectable Byzantine agreement. However, the protocol assumes that no more than one-third of the generals are disloyal. This threshold is standard in Byzantine agreement but limits the protocol's applicability in environments with high adversarial participation. Additionally, assuming TP to be semi-honest may be a security risk if the TP becomes malicious. This dependency may not be aligned with fully decentralized and trustless systems, such as certain blockchain applications.

Banerjee et al. [1] proposed a quantum blockchain framework that replaces classical blockchain cryptographic hash functions

with the entanglement of weighted hypergraph states. In this system, classical block data are encoded into the relative phases of quantum states, which are represented as vertices of a hypergraph. The entanglement among these quantum states forms the "chain" in the blockchain. Each block's information is encoded as a unique phase within a quantum state, using agreed-upon rules among peers, ensuring consistency across the blockchain. The chain is constructed incrementally, with peers verifying each block using measurements of the quantum states before adding it to their local copy of the blockchain. The consensus process is simplified, as it leverages quantum mechanics instead of classical cryptographic tools like PoW or PoS. Peers verify the authenticity of new blocks by checking whether their quantum properties adhere to the pre-defined consensus rules.

The security of the system relies on the entanglement properties of the quantum hypergraph states, which prevents tampering with the chain without destroying the entanglement and collapsing the state. This mechanism eliminates the need for computationally expensive cryptographic hash functions, ensuring both cost-effectiveness and security against classical and quantum computational threats. However, scalability remains limited to the capacity of current quantum technology, and the protocol is presented as a conceptual step forward rather than an immediate practical solution.

Wen et al. [38] utilized Quantum Zero-Knowledge Proofs (QZKPs) that leverage quantum states and the principles of quantum mechanics to achieve zero-knowledge properties in their consensus protocol. In QZKPs, the prover encodes the secret information or proof into a quantum state (e.g., a superposition or entangled state) that the verifier cannot directly measure without disturbing it. The verifier sends a quantum challenge (e.g., a specific measurement basis or entanglement test) to the prover. The prover processes the quantum state according to the challenge, using their secret, and sends back a quantum response. The verifier measures the returned quantum state and checks if it satisfies the expected outcomes. The quantum properties, such as entanglement or phase relations, ensure that the response is valid only if the prover has the correct secret. The no-cloning theorem and quantum uncertainty ensure that the verifier learns nothing beyond the validity of the proof because the quantum states cannot be copied or fully measured without losing information. In this consensus protocol, the user generates a random value by measuring a sequence of photons in a superposition state. This value serves as the basis for the mining. Miners generate their random value and perform a quantum zero-knowledge proof to validate their random value against the user's. If the miner's random value matches the user's, the miner passes the verification and generates a legitimate block. The block is sent to the network for validation. Upon successful validation by other miners, the miner who generated the block earns the right to keep the block in the chain and is rewarded. The network synchronizes with the new block and the miners proceed to compete for the next block. The authors argue that their proposed protocol exhibits the following performance properties, energy efficiency, high throughput, and shorter time delays. However, reliance on quantum infrastructure (e.g., quantum key distribution and photon

generation) poses significant technical and cost barriers. Additionally, the protocol depends on ideal quantum systems, which may be affected by noise and errors in real-world implementations.

5 Concluding Remarks

We observed that not all proposed protocols have accompanying security proofs that formally evaluate their security (see Table 4). A number of studies do not clearly describe their security definitions. For example, Shoup [33] pointed out that Dolev et al. [12] informally defined a secrecy property, but such a definition is not incorporated in the security proof and the assumptions needed for its proof is also not explained. We also observe this similar concerning trend in the reviewed studies. Thus, there should be an emphasis on having a robust and rigor security evaluation in future protocol design.

Also worth mentioning is that most post-quantum protocols assume a theoretical rigor in relying on an NP-hard problem for their security. However, while NP-hard problems generally have complex mathematical structure, they are only hard in the worst case. This may not be sufficient in many real-world applications. Crypto-systems need to be hard on average so that every instance is secure, and only a small number of problems (e.g. LWE) enjoy formal worst-to-average case reduction. As proven by Regev [31], LWE enjoys a worst-to-average case reduction from lattice problems such as Gap-SVP and SIVP which are believed to be hard.

One surprising observation is the lack of consensus protocols based on isogeny or code theory. This could be a promising direction for future research.

Although designing secure quantum-resistant protocols can potentially mitigate some of the risks associated with quantum computers, there are a number of challenges. For example, *how do we address adoption barriers such as the cost associated with migrating existing / legacy systems to quantum-ready / quantum-resistant ones?* It is also observed that most protocols did not implement their proposed consensus algorithms, which raises questions regarding their readiness for real-world deployment. A proposed quantum protocol may experience more implementation-related challenges as compared to post-quantum protocols, partly due to the need for expensive equipment and devices such as qubits and quantum channels. Thus, it is also important that existing and future protocols be tested for performance and scalability, particularly in real-world settings. Table 6 provides a summary of the limitations aforementioned.

It is important to acknowledge certain limitations of this review. While every effort was made to provide a comprehensive account of quantum-resistant consensus protocols, some relevant works may have been unintentionally omitted. Moreover, this survey primarily focuses on conceptual and security aspects, without an in-depth evaluation of practical implementation or performance comparisons. Future research could address these gaps by incorporating a unified set of comparison metrics and offering a more thorough discussion of real-world deployment challenges.

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Table 6: Summary of the Identified Limitations

Limitation	Implication
Major lack of security proofs	Such protocol may not be secure
Major reliance on worst case complexity	Some instance of the cryptographic problems used by such protocols may be easy to solve
Lack of discussion on adoption or migration to quantum-resistant systems	May hinder timely adoption of these protocols, resulting in higher costs and operational disruptions
Major lack of implementation	Such protocols may not be ready for real world deployment
Costly equipment required for quantum protocols.	Such protocols may face higher implementation challenges in real-world deployment

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